MICROSERVICES COURSE

Session 6

Inter-Service Communication

Sync vs Async · OpenFeign · Design Choices · JWT Propagation

Dr. ElSayed Mohamed Elsayed Baladoh

Tech Lead · Ph.D. · sayedbaladoh@yahoo.com

Monday

3:00 PM – 5:30 PM

Online Session

2.5 Hours

Phase 1

Session 6 of 29

WHERE WE ARE

Phase 1 — Foundation & Core Patterns

S1	S2	S3	S4	S5	S6	S7	S8
Architecture	Gateway Core	Gateway Adv.	Resilience CB	Resilience BH	OpenFeign	Saga+Kafka	Caching+Clinic
					● LIVE		

You are here — Session 6 of 8 in Phase 1. The platform survives failure. Today services finally TALK to each other.

BEFORE WE WRITE ANY CODE

This Session Is Not About OpenFeign

It is about how engineers DECIDE communication patterns.

The real skill is not writing @FeignClient — it is answering: "Why did I choose this pattern, and not another one?"

Sync or Async?

Does the caller need the answer to continue?

Feign, WebClient, or events?

Blocking call, reactive stream, or fire-and-forget?

What if it fails?

Retry? Compensate? Accept eventual consistency?

Sessions 6 & 7 are where you move from "how do I use the tool" to "how do I design the architecture."

PROBLEM FIRST

Order Service Doesn't Actually Check Inventory

Right now: createOrder() goes straight to Payment. There is no Inventory Service. Nothing stops an order for a product with ZERO stock.

1. Customer orders 5 units of PROD-003
2. PROD-003 has 0 units in stock
3. Order Service has no way to know this
4. **Payment succeeds. Order is CONFIRMED.**
5. **Warehouse has nothing to ship. Customer is angry.**

Today: build Inventory Service, and connect it to Order Service — synchronously, with intent.

DESIGN CHOICE

Sync or Async? Three Questions

Q1

Does the caller need the answer to CONTINUE?

YES → Sync (Feign/WebClient)
NO → Async (Kafka, Session 7)

Q2

Is the operation reversible if it fails?

YES → Sync is fine, retry
NO → Saga + compensation (S7)

Q3

Is eventual consistency acceptable?

NO → Sync (check stock NOW)
YES → Async (notify later)

Platform mapping:

Order → Inventory (check stock)	SYNC
Order → Notification (send email)	ASYNC
Order → Payment (charge card)	SYNC

DESIGN CHOICE

RestTemplate vs OpenFeign vs WebClient

DESIGN CHOICE

Simple synchronous call, need it now? → **OpenFeign**

High-throughput reactive pipeline? → **WebClient**

Starting a new project today? → **Avoid RestTemplate (deprecated path)**

	RestTemplate	OpenFeign	WebClient
Style	Imperative	Declarative interface	Reactive / functional
Blocking	Yes	Yes	No
Status	Deprecated	Recommended (sync)	Recommended (reactive)

LIVE CODING

Building Inventory Service

StockItem.java + InventoryService.java

```
public record StockItem(String productId, int availableQty, int reservedQty) {
    public boolean hasStock(int requested) { return availableQty - reservedQty >= requested; }
}
// In-memory: PROD-001 (100 units), PROD-002 (5 units), PROD-003 (0 units — out of stock)
```

InventoryController.java — GET /api/v1/inventory/check

```
@GetMapping("/check")
public ResponseEntity<StockCheckResponse> checkStock(@RequestParam String productId, @RequestParam int quantity)
{
    var response = inventoryService.checkStock(productId, quantity);
    return response.available() ? ResponseEntity.ok(response)
                                : ResponseEntity.status(409).body(response);
}
```

DEMO

Testing Inventory Service Directly

```
GET .../check?productId=PROD-001&quantity=5
```

200 OK *available: true, remainingStock: 100*

```
GET .../check?productId=PROD-003&quantity=1
```

409 Conflict *available: false, remainingStock: 0*

Inventory Service works correctly on its own. Next: Order Service calls it — synchronously, with OpenFeign.

LIVE CODING

Declaring the Feign Client

OrderServiceApplication.java + InventoryClient.java

```
@SpringBootApplication
@EnableFeignClients   // REQUIRED — scans for @FeignClient interfaces
public class OrderServiceApplication { ... }

@FeignClient(name = "INVENTORY-SERVICE", path = "/api/v1/inventory")
public interface InventoryClient {
    @GetMapping("/check")
    StockCheckResponse checkStock(@RequestParam("productId") String productId,
                                  @RequestParam("quantity") int quantity);
}
```

name = "INVENTORY-SERVICE" must match the Eureka service name EXACTLY. Feign resolves the address via Eureka — no hardcoded URL anywhere.

LIVE CODING

Calling Inventory Before Payment

OrderService.java — createOrderAsync()

```
@Bulkhead(...) @TimeLimiter(...) @CircuitBreaker(...) @Retry(...)
public CompletableFuture<OrderResponse> createOrderAsync(OrderRequest request) {
    return CompletableFuture.supplyAsync(() -> {
        // Step 1: Check inventory BEFORE payment
        StockCheckResponse stock = inventoryClient.checkStock(
            request.getProductId(), request.getQuantity());
        if (!stock.available()) {
            return new OrderResponse("REJECTED",
                "Insufficient stock: only " + stock.remainingStock() + " available");
        }
        // Step 2: Process payment (only if stock is OK)
        PaymentResponse payment = paymentService.processPayment(
            new PaymentRequest(request.getAmount()));
        return new OrderResponse("CONFIRMED", payment.getTransactionId());
    });
}
```

DEMO

The Full Chain — Order Checks Inventory First

POST /api/orders

`productId: PROD-003, quantity: 1`

↓

Order Service → Inventory Service

`Feign call via Eureka`

↓

Inventory responds

`409 — out of stock`

↓

Order Service response

`REJECTED — "Insufficient stock: only 0 available"`

Try PROD-001 instead → CONFIRMED. Watch both services' logs to see the cross-service call.

LIVE CODING

Translating HTTP Errors into Domain Exceptions

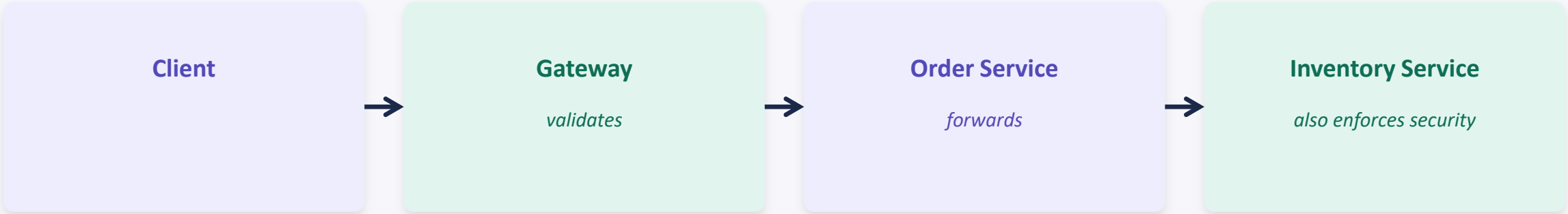
InventoryErrorDecoder.java

```
@Component
public class InventoryErrorDecoder implements ErrorDecoder {
    @Override
    public Exception decode(String methodKey, Response response) {
        return switch (response.status()) {
            case 409 -> new InsufficientStockException("Product out of stock");
            case 404 -> new ProductNotFoundException("Product not found");
            case 503 -> new ServiceUnavailableException("Inventory unavailable");
            default  -> new FeignException.InternalServerError(
                "Unexpected error", null, null, null);
        };
    }
}
```

By default, Feign wraps ALL HTTP errors as generic FeignException. ErrorDecoder translates them into meaningful domain exceptions your business logic can handle.

CORE CONCEPT

JWT Propagation — The Trust Chain Continues



The JWT travels with the request through the ENTIRE service chain.

Gateway validates it once. Order Service reads it. Order forwards it to Inventory via a RequestInterceptor. Inventory can independently check who the user is.

LIVE CODING

Propagating the JWT to Inventory Service

FeignJwtInterceptor.java — Order Service

```
@Component
public class FeignJwtInterceptor implements RequestInterceptor {
    @Override
    public void apply(RequestTemplate template) {
        var attrs = (ServletRequestAttributes) RequestContextHolder.getRequestAttributes();
        if (attrs != null) {
            String authHeader = attrs.getRequest().getHeader(HttpHeaders.AUTHORIZATION);
            if (authHeader != null) {
                template.header(HttpHeaders.AUTHORIZATION, authHeader);
            }
        }
    }
}
// Applies to ALL Feign clients in Order Service automatically
```

DESIGN CHOICE

WebClient — When Would You Reach for It?

Use WebClient when:

- Your service is built on WebFlux (reactive) throughout
- You need streaming responses (SSE, chunked data)
- You need fine-grained per-call control (custom timeouts, codecs)

In Our Platform Today:

OpenFeign for ALL service-to-service calls — blocking, simpler, matches our Spring MVC stack.

WebClient is used internally by Spring Cloud Gateway (it is WebFlux-based) — but we don't write it ourselves.

WARNING: calling .block() inside a WebClient chain defeats its entire purpose — never do it in production code.

HANDS-ON

Lab 4A — Inventory Service + OpenFeign Order Call

1	**Build Inventory Service**	New service, port 8084, in-memory stock with 3 products
2	**Add OpenFeign Client to Order Service**	InventoryClient interface, check stock before payment
3	**JWT Propagation**	FeignJwtInterceptor forwards Authorization header
4	**Write Unit Tests**	Minimum 3 tests covering Feign call + error handling

DEFINITION OF DONE

Lab 4A Acceptance Criteria

- [x] GET /api/v1/inventory/check?productId=PROD-001&quantity=5 → 200, available:true
- [x] GET .../check?productId=PROD-003&quantity=1 → 409, available:false
- [x] POST /api/orders with PROD-003 → REJECTED (insufficient stock)
- [x] POST /api/orders with PROD-001 → CONFIRMED (stock checked + payment processed)
- [x] JWT forwarded to Inventory Service — visible in Inventory's logs
- [x] ErrorDecoder translates 409 into InsufficientStockException
- [x] All 3 unit tests pass: mvn test
- [x] Commit: session-06: add-inventory-service-and-feign-client

PLATFORM PROGRESS

What We Added to the Platform Today

Capability Added

Sync Service-to-Service Communication

Services now talk via typed interfaces. Order checks real inventory before confirming — business rules cross service boundaries cleanly.

Services Added / Updated

- NEW: Inventory Service (8084)
- UPDATED: Order Service — InventoryClient (Feign)
- FeignJwtInterceptor — JWT propagation
- InventoryErrorDecoder — domain errors

- Sessions 1-5: Config + Eureka + Gateway + Payment + Order (full resilience)
- Session 6: Inventory Service + OpenFeign — first real service-to-service call

Config	Eureka	Product	Gateway Sec.	Payment	Order Resil.	Inventory	Notif. S7

Daily Quiz

8 Questions · 10 Minutes

Topics: Sync vs Async Decision · OpenFeign vs WebClient · JWT Propagation · ErrorDecoder

BEFORE SESSION 7

Checkpoint Commit & Homework

Checkpoint — Definition of Done

- ☐ Inventory Service running on :8084
- ☐ Order rejects PROD-003 (no stock)
- ☐ Order confirms PROD-001 (has stock)
- ☐ JWT visible in Inventory logs
- ☐ Pushed: session-06: ...

Pre-Session 7 Reading (15 min)

Saga Pattern + Distributed Transactions
microservices.io/patterns/data/saga.html

Knowledge Check Questions for Session 7:

- What is Choreography Saga vs Orchestration Saga?
- What happens if Payment succeeds but Inventory fails?
- What is a Kafka consumer group?

TROUBLESHOOTING

Common Issues & Solutions

Feign client returns "no instances available"	Inventory Service must be UP and registered in Eureka before Order Service starts calling it
@FeignClient interface has no implementation	Missing @EnableFeignClients on the main application class
ErrorDecoder not catching errors	Register as @Component — Spring must discover it as a bean to wire into Feign
JWT not appearing in Inventory logs	Check FeignJwtInterceptor is a @Component and Authorization header exists on the original request
StockCheckResponse fields don't match	Record fields must match Inventory's response exactly — duplicate the record carefully

Session 6 Complete

Services now talk to each other — synchronously, with intent

Next: Session 7 — Distributed Transactions: Saga Pattern + Kafka · Wednesday, 3:00–5:30 PM, Online

microservices-pro-course · microservices-pro-platform · Dr. Sayed Baladoh